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The Jesus DNA Hypothesis: A Quantitative Reassessment of Patrilineal Survival and Carrier Numbers for a 1st-Century Judean Founder

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ABSTRACT

Background. *The Jesus DNA Hypothesis* is a one-page theoretical model asking what would survive today of one 1st-century Judean man's uniparental DNA. It concludes that, if his male line survived, about 200,000 living men would carry his Y chromosome.

Methods. Each quantitative claim was checked against the primary literature. The survival and size of a single founder's male line were modelled as a Galton–Watson branching process, with 2 million simulated founders per scenario across 65 or 80 generations, 1- to 30-fold population growth and two levels of variance in sons per man. Marker mutation counts were recomputed from pedigree-based rates.

Results. The model's extinction curve is accurate: about 63%, 16% and 9% of lines survive 1, 10 and 20 generations, and 2–10% survive 65–80. The carrier estimate is not. A surviving line reaches a median of 24–205 living men, and none of 10 million simulated founders exceeded 4,000. Reaching 200,000 requires reproductive advantage on the scale of empire-driven lineages, which the hypothesis itself excludes. Several marker figures need minor correction, chiefly the Y-chromosome length (~62.5 Mb), the generation count (~64–65) and the expected mutations (~109).

Conclusions. The hypothesis rests on a sound mechanism but overstates the magnitude by roughly three to four orders. If the line survived, it is most likely carried by a few hundred living men. Model code is released for reproducibility.

Keywords · Y chromosome · mitochondrial DNA · uniparental markers · lineage extinction · Galton–Watson branching process · genetic genealogy · Levant · science communication

1 Introduction

This paper reassesses *The Jesus DNA Hypothesis* (hereafter **the sheet**), a one-page theoretical model of what a single 1st-century Judean man's DNA would leave in living people after about 2,000 years. The sheet is explicit that it is a mathematical exercise rather than a claim about any person's descendants, and this reassessment keeps that framing.

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The sheet argues in three steps:

1. **The Y chromosome is a receipt.** It passes from father to son almost unchanged, so after 80 generations it still matches the founder's at 99.99998% of sites (panel 01).
2. **Most male lines die out.** The founder's line has a 95–97% chance of being extinct today (panel 02).
3. **A surviving line is large.** If it survived, about 200,000 living men would carry it, within a range of 50,000 to 1 million, concentrated among Maronite, Orthodox and Druze communities in Lebanon, Samaritan and Christian communities in Israel and Palestine, and populations in Syria and Jordan (panels 03 and 05).

The sheet also covers maternal inheritance through mtDNA and the shared autosomal genome (panel 04), and compares the hypothetical line with known Y-chromosome expansions (panel 06).

Three distinctions frame the assessment:

1. **Genealogical versus genetic ancestry.** Anyone alive 2,000 years ago who left descendants is probably a genealogical ancestor of a very large share of people living today (Rohde, Olson & Chang 2004). Carrying that person's Y chromosome is far rarer, because it follows a single path through the family tree.
2. **Haplogroup versus individual.** Sharing a haplogroup such as J1 shows deep common ancestry, usually thousands of years older than any named person. It does not identify an individual founder.
3. **Tested versus inferred descent.** No Y-chromosome data exist for any 1st-century individual relevant to the sheet, so every carrier figure is a model output, not an observation.

This paper makes no religious or theological claim. In practice, the Y line in question would be Joseph's patriline, carried forward through the brothers named in Mark 6:3 under traditions that treat them as Joseph's sons; the founder is treated here only as a hypothetical 1st-century Judean man.

2 Materials and methods

The reassessment combined a claim-by-claim literature check, a branching-process model of patrilineal survival and size, and recomputed mutation arithmetic.

Material. The sheet: a single page in six numbered panels (01 the Y chromosome; 02 line survival; 03 carrier numbers; 04 maternal and autosomal inheritance; 05 geographic distribution; 06 comparison with known Y-line expansions), with its stated sources.

Literature check. Each quantitative or factual claim was traced to the stated source where one was given, and to the primary literature where it was not. Claims were graded *Supported*, *Needs correction*, or *Unsupported* (no published basis located).

Lineage model. Survival of a single founder's male line was modelled as a Galton–Watson branching process, in which each man has an independent random number of sons. Survival probabilities were computed exactly by iterating the offspring generating function. Lineage sizes were estimated from 2,000,000 simulated founders per scenario (seed 20261111). Scenarios varied three inputs:

- **Generations:** 80 (the sheet's implied 25-year interval) and 65 (a 31-year male generation interval over the 1,996 years from 30 AD to 2026).

- **Population growth over the period:** none, 10-fold and 30-fold, applied as a constant per-generation mean number of sons.
- **Variance in sons per man:** Poisson (variance equal to mean) and overdispersed negative binomial (shape $k = 1$, variance about twice the mean).

Mutation arithmetic. Expected point mutations along one 1,996-year patriline used the pedigree-based Y rate of 8.71×10^{-10} per site per year (Helgason et al. 2015) and the complete Y length of about 62.5 Mb (Rhie et al. 2023). The mtDNA comparison used 1.665×10^{-8} per site per year over 16,569 bp (Soares et al. 2009).

Figures. Figures 1 and 3 are drawn directly from the model output. Figures 2 and 4 redraw the sheet's panels 03 and 05 from the values printed on it, beside the model's results. Figure 5 combines the model output with published carrier estimates for known Y-chromosome expansions.

3 Background: the markers

The sheet's central image (panel 01), a near-perfect Y-chromosome "receipt" after two millennia, holds. Two properties of uniparental markers drive everything that follows.

They change slowly, but not zero. A patriline differs from its founder only by new mutations. At 8.71×10^{-10} per site per year (Helgason et al. 2015), about 109 point mutations accumulate across the ~62.5 Mb Y chromosome in 1,996 years, leaving ~99.9998% identity. Mitochondrial DNA mutates about 19 times faster per site (Soares et al. 2009), but the molecule is roughly 3,800 times smaller, so a maternal line gains only ~0.55 differences in the same period. Y short-tandem repeats (STRs) mutate far faster than point sites, so STR haplotypes shift within a few generations.

Identity is not identification. Every patriline is near-identical to its own founder, and two unrelated men from the same region typically differ at only hundreds to a few thousand Y positions. A living man's Y chromosome can therefore be linked to a particular 1st-century founder only through a sample from that founder or a well-documented genealogy, neither of which exists here. This is why every carrier figure in the sheet, and in this paper, is a model output.

4 Results: survival and carrier numbers

The sheet's extinction curve is broadly right; its carrier count is not. A surviving 2,000-year patriline reaches tens to a few thousand living men under neutral inheritance, roughly 50 to 8,000 times fewer than the sheet's central estimate of ~200,000.

4.1 Line survival (panel 02)

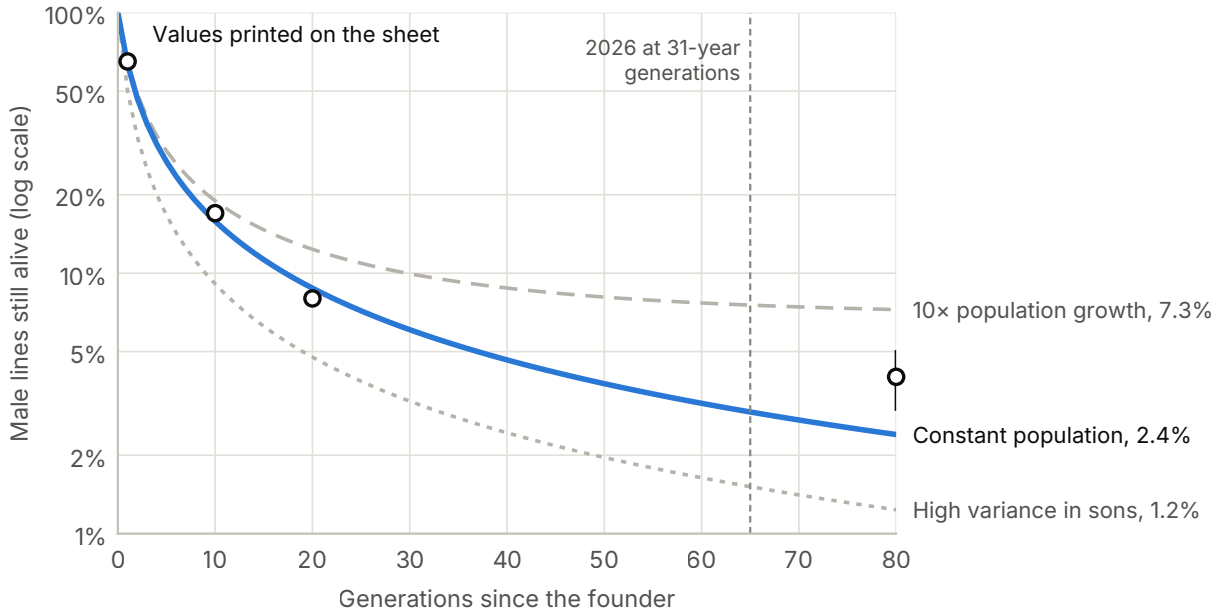


Figure 1. The sheet's extinction curve tracks a neutral branching model. Share of founders whose male line is still alive, by generation (log scale). Lines show exact survival probabilities from the Galton–Watson model; the per-generation growth rate is set for 65 generations and extended to 80. Open circles are the values printed on the sheet (panel 02); the bar at generation 80 spans its 3–5% range. The dashed vertical line marks the present at a 31-year generation interval.

The model reproduces the sheet's printed values within two percentage points at generations 1, 10 and 20. At 65 generations, 1.5–7.5% of lines survive depending on growth and variance, so the sheet's 3–5% and its "95–97% extinct" are reasonable central statements. One refinement: the sheet's premise stipulates a founder who had a son. Conditioning on that first step raises survival to about 3–12% at 65 generations.

4.2 Carrier numbers (panel 03)

Panel 03 gives three scenarios for a surviving line: a conservative 50,000–100,000 men if descendants stayed in Galilee villages with little branching; a moderate 200,000–500,000, the sheet's preferred case, with spread across the Levant and the modern diaspora; and an upper bound near 1 million with strong branching and Ottoman-era and colonial migration. Its headline figures, ~200,000 men or about 1 in 20,000 men worldwide (0.005%), sit at the low end of the moderate case. Figure 2 sets all three beside the model's result.

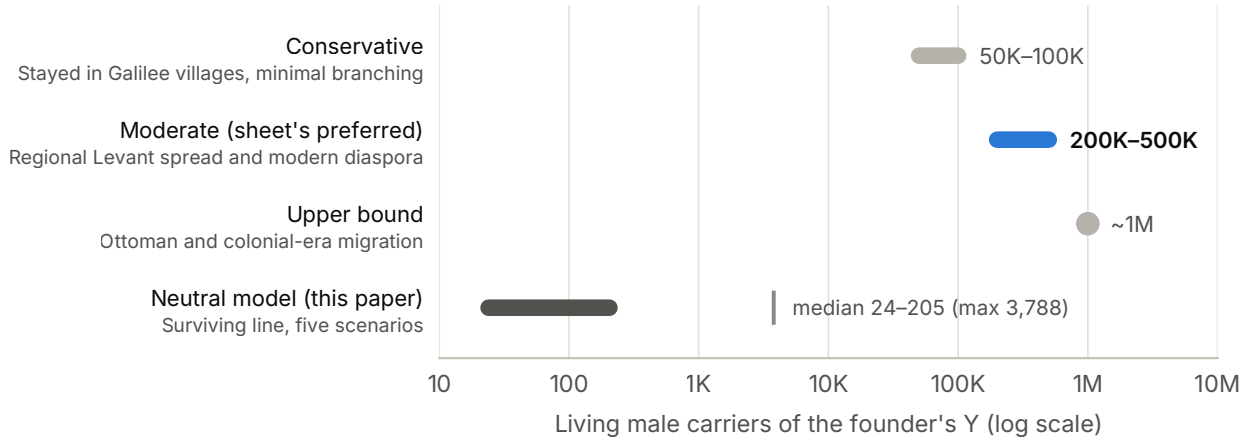


Figure 2. Even the sheet's conservative scenario lies far above a neutral line. The three carrier scenarios printed in panel 03, with the conditions the sheet attaches to each, compared with the range of median surviving-line sizes across this paper's five model scenarios (tick: largest of 10 million simulated founders). Log scale.

The model shows why the gap exists (Figure 3).



Figure 3. Surviving male lines reach hundreds of men, not 200,000. Living men carrying one founder's Y chromosome after 65 generations, among lines that survived. Dots mark the median, bars extend to the 99th percentile, and ticks mark the largest of 2,000,000 simulated founders per scenario (seed 20261111). The dashed line marks the sheet's central estimate.

The arithmetic behind Figure 3 is short. Unconditionally, a founder's expected number of male-line male descendants equals the population's growth factor: 1 in a stable population, 10 after 10-fold growth. Dividing by the survival probability gives the expected size of a surviving line, about 34–290 men across the scenarios. None of 10 million simulated founders reached 4,000 carriers, so even the sheet's "conservative" 50,000 is more than 13 times the largest simulated line.

4.3 Where carriers would live (panel 05)

Panel 05 assigns 30–40% of carriers to Lebanon, among its Maronite, Orthodox and Druze communities; 25–35% to Israel and Palestine, among Samaritan and Christian communities; 10–15% to Syria and Jordan; 10–20% to the Levantine diaspora in the Americas, Australia and Europe; and under 5% to the rest of the world (Figure 4). It adds that within any single community 5–15% of men would carry the line, rising to 25–30% in small isolated villages and falling to 1–2% in cities such as Beirut.

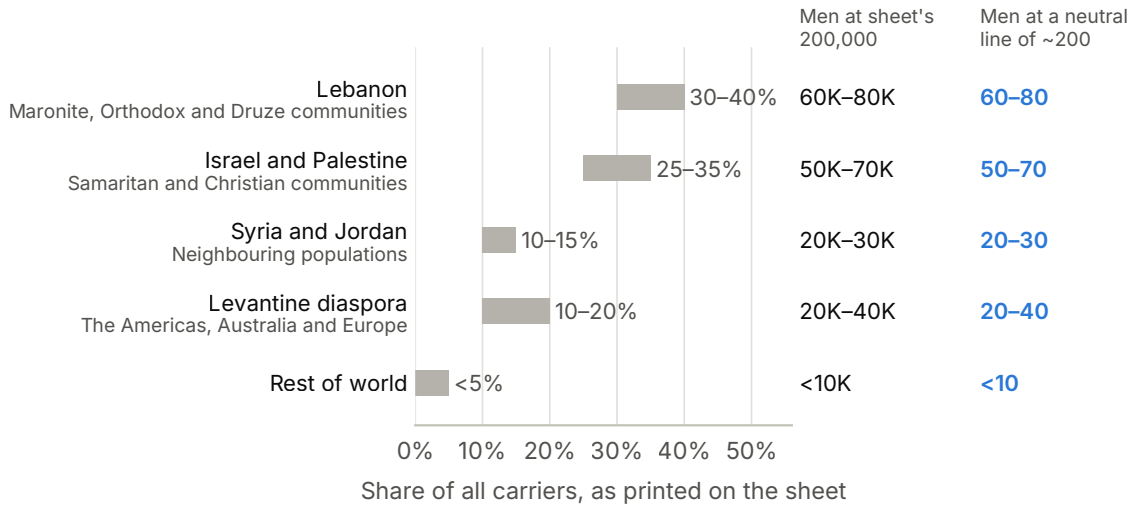


Figure 4. The sheet's geographic shares, and the numbers of men they imply. Bars show the share of all carriers that panel 05 assigns to each region, with the communities it names. The right-hand columns convert those shares into men at the sheet's total of 200,000 and at a neutral surviving line of about 200 men (Figure 3).

No model or source in the sheet produces these shares; they are assumptions. They also drive the headline total. A 5–15% share of men across the named communities, which together number in the hundreds of thousands, already implies tens of thousands of carriers in Lebanon alone, so the large total is an input to the estimate rather than an output of any inheritance process the sheet describes. Applied to a neutral line of about 200 men, the same percentages would mean roughly 60–80 men in Lebanon and 50–70 in Israel and Palestine, scattered across communities rather than forming a measurable share of any of them.

4.4 Comparison with known expansions (panel 06)

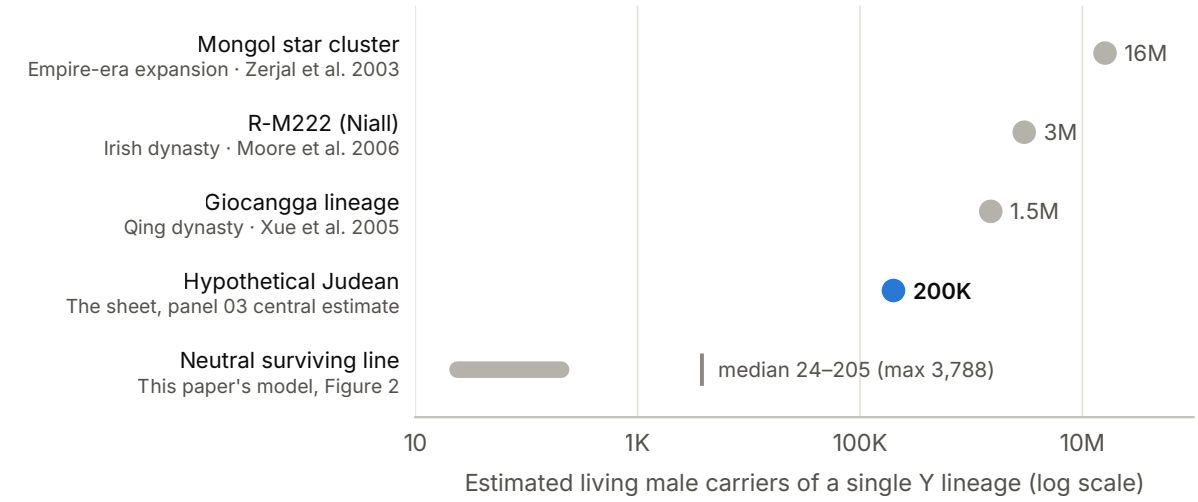


Figure 5. 200,000 carriers would place a founder among dynasty-driven lineages. Estimated living male carriers of single Y lineages (log scale). Published estimates: Zerjal et al. (2003), Moore et al. (2006), Xue et al. (2005). The sheet's value is its panel 03 central estimate; the neutral range is the span of median line sizes across the scenarios in Figure 3, with the tick marking the largest simulated line.

Panel 06 correctly identifies what made the known expansions large: empire, dynasty and the reproductive advantage that came with them. Those are exactly the conditions a 200,000-man line would require, and the sheet's founder had none of them. Two of the comparison lineages are also now weaker anchors than the sheet suggests. The Mongol star cluster has been traced to ordinary Mongol tribes with a common ancestor about 2,576 years ago rather than to Genghis Khan (Wei et al. 2018), and the link between R-M222 and Niall remains debated.

The panel's final row, "Average man, any era: 0 carriers", is too strong. About 2–10% of lines survive, so the average founder leaves close to zero carriers, but a typical surviving line leaves hundreds.

5 Results: panel-by-panel review

Of 15 checkable claims on the sheet, 4 are supported, 9 need correction and 2 have no published basis. Most corrections are small; the two that change the conclusion are the carrier count (panel 03) and the geographic shares that feed it (panel 05).

Table 1. Claims on the sheet, findings and verdicts.

Panel	Claim on the sheet	Finding	Verdict
01	99.99998% match to the original after 2,000 years	~99.9998% (about 109 differences in 62.5 Mb); still a near-perfect match	Needs correction
01	Y is ~57 million base pairs, 0.9% of a man's genome	~62.5 Mb in the complete sequence (Rhie et al. 2023); about 1%	Needs correction
01	~13–14 mutations over 80 generations	~109 across the full Y; ~26 in the 15.2 Mb measured by Helgason et al. (2015)	Needs correction
01	80 generations	~64–65 at a 31-year male generation interval; 80 assumes 25 years	Needs correction
01	NRY is ~95% of the Y and acts as a non-coding barcode	~95% is correct; the region is mostly non-coding but carries dozens of protein-coding genes besides SRY and AZF	Supported
01	Likely haplogroup J1 or J2	A regional prior, not an inference: Levantine men today also carry E1b1b, T, G, R1b and others, and no 1st-century Y data exist for the founder	Unsupported
02	95–97% chance the line is extinct; ~65%, ~17%, ~8% survive at generations 1, 10, 20	Matches the model within two percentage points (Figure 1)	Supported
02	A line that survives "the gauntlet" branches enough to become permanent	Surviving lines stay small (Figure 3) and keep facing extinction; in a stable population survival keeps falling roughly as $2/n$	Needs correction
03	~200,000 living carriers (50,000–1 million); 1 in 20,000 men	Neutral median 24–205; largest of 10 million simulated lines 3,788 (Figures 2 and 3)	Needs correction
04	Mary's mtDNA would be a 99.999%+ match after 80 generations	~0.55 expected differences in 16,569 bp; about 58% of lines identical, the rest ~99.994%	Needs correction
04	Daughters' daughters would carry the maternal line unchanged	Correct in principle; requires an unbroken chain of daughters, and no data exist	Supported
04	Endogamous Levantine communities carry ~93% Canaanite ancestry (Haber et al. 2017)	That estimate is for present-day Lebanese genomes modelled against Bronze Age Sidon; it does not extend to all Levantine groups or describe a 1st-century Judean	Needs correction

Panel	Claim on the sheet	Finding	Verdict
05	Carriers 30–40% in Lebanon (Maronite, Orthodox, Druze), 25–35% in Israel and Palestine (Samaritan, Christian), 10–15% Syria and Jordan, 10–20% diaspora; 5–15% of men within communities, 25–30% in villages	An assumed distribution with no model or source behind it; incompatible with neutral line sizes (Figure 4)	Unsupported
06	Genghis Khan ~16M, Giocangga ~1.5M, Niall ~3M carriers	Match the published estimates (Zerjal et al. 2003; Xue et al. 2005; Moore et al. 2006), though attribution to the named men is contested (Wei et al. 2018)	Supported
06	Average man, any era: 0 carriers	~2–10% of lines survive; the average is near zero, but a typical surviving line has hundreds of carriers	Needs correction

6 Discussion and recommended revisions

The sheet teaches the right mechanism and reaches the wrong magnitude. The Y chromosome is a near-perfect record and most male lines do go extinct; but a line that survives without social advantage stays small, and the famous large lineages are large precisely because they had that advantage.

The revisions below are ordered by how much they change what a reader concludes.

1. **Replace the ~200,000 carrier estimate** (panel 03) with the neutral range: a median of roughly 25–200 living men, rarely more than a few thousand. Present 200,000 or more as the size that would require dynasty-scale reproductive advantage. The "1 in 20,000 men" figure becomes roughly 1 in 10–100 million.
2. **Remove or relabel panel 05.** The country shares and the 5–15% community share are assumptions, not results. A defensible statement is qualitative: carriers, if any, would most likely live where the founder's descendants stayed, with a minority in the diaspora.
3. **Revise panel 02's closing claim.** Surviving the first 20 generations does not make a line permanent; it continues to face extinction.
4. **Correct panel 01's figures:** ~62.5 Mb, ~64–65 generations, ~109 mutations and ~99.9998% identity.
5. **Present J1 or J2 as a regional prior**, not a likely haplogroup for the founder.
6. **Narrow the Haber et al. (2017) citation** to present-day Lebanese, and drop it as a description of a 1st-century Judean.
7. **Adjust the mtDNA identity** to ~99.997% expected, noting that most maternal lines would be identical.
8. **Add caveats to panel 06:** the Genghis Khan and Niall attributions are contested, and the "Average man: 0" row should distinguish the average founder from a typical surviving line.
9. **Expand the source list** to include Helgason et al. (2015), Rhie et al. (2023), Soares et al. (2009), Wei et al. (2018) and Xue et al. (2005).

With these changes the sheet's headline becomes both accurate and still striking. A 1st-century Judean patriline probably did not survive; if it did, it is most likely carried today by a few hundred men, each with a Y chromosome 99.9998% identical to the founder's after two thousand years.

7 Limitations, ethics and availability

Model limitations. The Galton–Watson model treats men as independent and the population as unstructured. Real populations add local density regulation, migration, endogamy and village-level drift, which can enlarge some lines locally; but none of these produces the several-thousand-fold excess needed for 200,000 carriers without differential reproductive success. Generation interval, growth factor and offspring variance were bracketed rather than estimated for the 1st-century Levant, and mutation counts use present-day pedigree rates.

Evidence limitations. No genetic data exist for the hypothetical founder, so this paper tests the sheet's internal logic and parameters, not any claim of actual descent. Published carrier estimates for comparison lineages (Figure 5) carry wide and in some cases contested uncertainty.

Ethical considerations. Claims of descent from religious figures carry deep personal, communal and religious significance. Numerical estimates of "carriers" can be read as genetic verdicts on particular families or communities. This paper, like the sheet, makes no claim that any living person or group descends from any historical figure, and its figures should not be used to support or dispute such claims. No personal or genetic data were collected.

Data and code availability. The lineage model (`lineage_survival_model.py`, Python 3 with NumPy, fixed seed 20261111) and its output (`lineage_survival_results.json`) accompany this paper in its Zenodo record. They reproduce every model-derived number in this paper, including those in Figures 1–5.

Funding and competing interests. None declared.

8 Statement on the use of AI assistance

A generative AI assistant was used to support literature retrieval, drafting and editing of the text, writing of the simulation code and preparation of the figures.

References

Haber, M., Doumet-Serhal, C., Scheib, C., et al. (2017). Continuity and admixture in the last five millennia of Levantine history from ancient Canaanite and present-day Lebanese genome sequences. *American Journal of Human Genetics*, 101(2), 274–282. doi:10.1016/j.ajhg.2017.06.013

Helgason, A., Einarsson, A. W., Guðmundsdóttir, V. B., et al. (2015). The Y-chromosome point mutation rate in humans. *Nature Genetics*, 47(5), 453–457. doi:10.1038/ng.3171

Moore, L. T., McEvoy, B., Cape, E., Simms, K., & Bradley, D. G. (2006). A Y-chromosome signature of hegemony in Gaelic Ireland. *American Journal of Human Genetics*, 78(2), 334–338. doi:10.1086/500055

Rhie, A., Nurk, S., Cechova, M., et al. (2023). The complete sequence of a human Y chromosome. *Nature*, 621, 344–354. doi:10.1038/s41586-023-06457-y

Rohde, D. L. T., Olson, S., & Chang, J. T. (2004). Modelling the recent common ancestry of all living humans. *Nature*, 431, 562–566. doi:10.1038/nature02842

Soares, P., Ermini, L., Thomson, N., et al. (2009). Correcting for purifying selection: an improved human mitochondrial molecular clock. *American Journal of Human Genetics*, 84(6), 740–759. doi:10.1016/j.ajhg.2009.05.001

Wei, L.-H., Yan, S., Lu, Y., et al. (2018). Whole-sequence analysis indicates that the Y chromosome C2*-Star Cluster traces back to ordinary Mongols, rather than Genghis Khan. *European Journal of Human Genetics*, 26(2), 230–237. doi:10.1038/s41431-017-0012-3

Xue, Y., Zerjal, T., Bao, W., et al. (2005). Recent spread of a Y-chromosome lineage in northern China and Mongolia. *American Journal of Human Genetics*, 77(6), 1112–1116. doi:10.1086/498583

Zerjal, T., Xue, Y., Bertorelle, G., et al. (2003). The genetic legacy of the Mongols. *American Journal of Human Genetics*, 72(3), 717–721. doi:10.1086/367774

The Gospel of Mark 6:3.